

ZACHARY RAUEN

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PROFILE

Multidisciplinary engineer with experience across firmware, systems design, web technologies, open-source development, and AI-augmented workflows. Proven ability to modernize legacy systems, lead large-scale projects, and deliver high-impact solutions across diverse technical domains.

PROFESSIONAL EXPERIENCE

Firmware Recovery Lead Engineer @ IBM – November 2021 - Present

- ❖ Lead engineer for sysplex recovery on IBM Z, designing and coordinating built-in recovery systems across mainframe firmware. Guided cross-team architecture decisions, established recovery-safe development practices, and delivered reliability improvements for business-critical system components.

Staff Firmware Engineer @ IBM – January 2019 - November 2021

- ❖ Developed core components of parallel sysplex using C/C++ and PL8, delivering high-availability features and direct-to-client debugging support across multiple squads.

HCI + Biometrics Researcher @ Clarkson University – August 2016 - May 2018

- ❖ Designed a crowdsensing-based continuous authentication framework; ran multi-session user studies, performed statistical analysis, and validated a novel gesture- and sociability-based verification approach.

Hardware & Software Developer @ IBM – Summer 2015

- ❖ Built a cross-site hardware recovery testing tool for IBM Z and Power systems, improving validation workflows.

Biometrics Researcher @ Clarkson University – Summer 2014

- ❖ Developed a keystroke-based user verification algorithm that improved accuracy from 80.2% to 96.8%. Reduced false acceptance rates from 12% to 2.73% and earned recognition at the SURE Conference.

PROJECTS

BetterDiscord – Open Source – github.com/BetterDiscord/BetterDiscord

- ❖ Created and led a widely adopted client mod for Discord, using React, Node, and reverse-engineering techniques to integrate with a closed-source platform. Grew the project to ~9,000 GitHub stars and over 50M downloads.

betterdiscord.app – Open Source – github.com/BetterDiscord/betterdiscord.app

- ❖ Rebuilt a legacy codebase into a modern TypeScript + SvelteKit + Bun + Drizzle stack using AI-assisted workflows. Migrated production within a month, serving ~100M monthly requests with more reliability and less hardware usage.

DumpXTR – IBM

- ❖ Revived a mainframe memory-dump extraction library by converting it into a modern Go-based ecosystem with CLI tools, language bindings, and WASM support. Used internal AI coding agents to accelerate the conversion process.

TECHNICAL SKILLS & TOOLING

Programming Languages

- C, C++, Go, TypeScript, C#, Java, Kotlin, Python

Web Technologies

- React, Svelte/Sveltekit, Node, Bun, Electron, Wails, Tailwind, PHP, WASM

AI-Augmented Dev

- Agentic Workflows, Automated Refactoring, Code Generation, Automated Maintenance

Systems & Firmware

- PL8, Assembly, Embedded & Recovery Systems

DevOps & Tools

- Linux, Bash/Zsh, Docker, Git, CI/CD, AHK, Lua, SQL

Hardware Design

- VHDL, Spice, FPGAs, Multisim, Xilinx, Quartus II, DSP

Research

- MATLAB, R, HCI, LaTeX, Biometrics, Experiments

EDUCATION AND TRAINING

Clarkson University 2018

Master of Science: Electrical and Computer Engineering **3.9 GPA**

Clarkson University 2016

Bachelor of Science: Computer Engineering **3.67 GPA**

AWARDS & ACHIEVEMENTS

IBM Client Care Award – July 2025

The client care award is given to developers that have gone above and beyond to deliver for clients in new features, debugging, or direct consultation. This award is driven by real feedback from customers.

IBM Tech 2023 Award – March 2023

Given to individuals with major contributions in advancing and modernizing the tech stack for their team or area. Winners have demonstrated the ability to lead by examples and accomplish real-world results.